



x

1

7

x

10

0

1

0

1



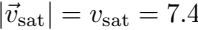




PARADOXES

ESPRESSO

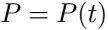








BEAR





THE FIRST TWO



100%







2023-10-21

$$= \underbrace{\frac{p_t^2}{2 \log(2)} \pi \lambda \left(\frac{h \varphi_{\text{RX}}}{\sin^2(\epsilon)} \right)^2}_{k \triangleq} \exp \left\{ - \tau^2 \underbrace{v_{\text{sat}}^2 \sin^4(\epsilon) / (h^2 \varphi_{\text{RX}}^2)}_{D \triangleq} \log(2) / 2 \right\},$$



























Fig 1(a): True $P(t)$, GPR estimate & forecast, and 100-sample MA & forecast

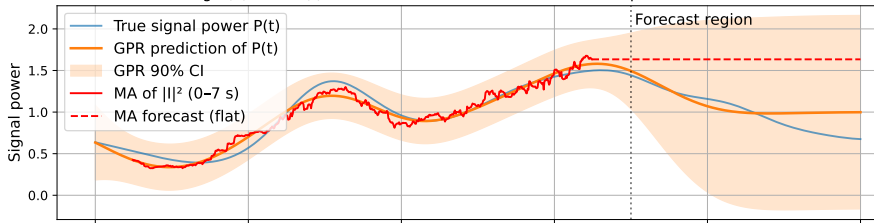
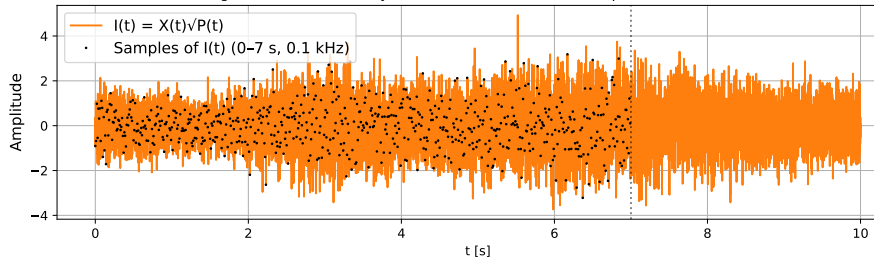
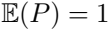


Fig 1(b): Non-stationary Gaussian noise $I(t)$ with sampled observations













expanding world + new world







Qatar Airways









$$S_{\text{RX}}(t) = \frac{A}{l(d_0)} \sqrt{\frac{K}{K+1}} e^{-2i\pi\tau_0(t)f_c} e^{-2i\pi\theta} + \sum_{j=1}^{100} \frac{A}{l(d_j) \sqrt{100} \sqrt{K+1}} e^{-2i\pi\tau_j(t)f_c} e^{-2i\pi\theta},$$











2020 = 2020







QWERTY

$$v = \sqrt{\frac{GM}{h + R_{\oplus}}},$$

WORLD OF









20

—

12

.

100



$$T_c = \frac{1}{8D_g} \approx 10^{-3}$$









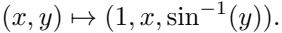


















$$1 = \frac{1}{\epsilon - \frac{\pi^2}{2\sqrt{P}}}$$





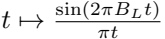
$$J = 1 - \left(1 - e^{-\frac{\pi^2}{2\sqrt{P/N}}} \right)^N,$$





192







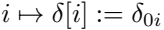
$$t \rightarrow \text{order} \rightarrow \frac{\sin(2\pi B t)}{\pi t}$$





$$\mathbb{Z} \ni i \mapsto \frac{\sin\left(2\pi \frac{B_L}{f_c} i\right)}{\pi i},$$

$$i \mapsto \delta[i] - \frac{\sin(2\pi \frac{B_H}{f_c} i)}{\pi i},$$



গণপ্রজাতন্ত্রী বাংলাদেশ

WAVELENGTH



WINNERS

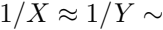
$$\mu = \exp\left(\mu_{\text{LN}} + \frac{\sigma_{\text{LN}}^2}{2}\right)$$

$\sin^2 \theta = 1 - \sin^2 \theta$









WORLD OF



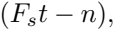
A 16x16 pixelated grayscale image of the letter 'A'. The image is composed of a grid of 256 pixels. The letter 'A' is rendered in a dark gray/black color, with some lighter gray pixels used for shading and detail. The background is white. The 'A' is formed by two main vertical strokes and a horizontal crossbar. The top of the 'A' is slightly wider than the bottom. The right side of the 'A' has a small, light gray pixelated detail near the top. The bottom of the 'A' is slightly wider than the top. The overall style is reminiscent of early digital art or computer graphics.

A 16x16 grayscale pixelated image of the number 75. The image is composed of a grid of 256 pixels, with varying shades of gray representing the digits. The number 7 is on the left and the number 5 is on the right. The style is reminiscent of early digital art or computer graphics.

Age Group	Total (%)	Male (%)	Female (%)	Non-binary (%)
18-24	15	12	18	10
25-34	25	22	28	15
35-44	20	18	22	12
45-54	18	16	20	10
55-64	12	10	14	8
65+	10	8	12	6

A pixelated, black and white representation of the number 20. The digit '2' is on the left, followed by an equals sign, and the digit '0' is on the right. The image has a low-resolution, dithered appearance.

$$S(t) = \sum_{n=0}^{N-1} x[n]$$













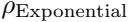


$$H = \begin{pmatrix} -a & -0 & -1 & 0 & 0 & 0 & ar & 0 & r \\ 0 & 0 & 0 & -a & -0 & -1 & 0 & 0 & 0 \\ -c & -b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & -b^2/a & -1 & cr & rb^2/a & r \\ a & 0 & -1 & 0 & 0 & 0 & ar & 0 & -r \\ 0 & 0 & 0 & a & 0 & -1 & 0 & 0 & 0 \\ -c & b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & b^2/a & -1 & -rc & rb^2/ar & -r \end{pmatrix}$$

$$\lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X \geq x)}{x},$$







$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{-x/\theta}{x} = 1/\theta.$$





$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(1 - \gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$\begin{aligned}
 &\stackrel{(a)}{=} \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta))}{x} = \lim_{x \rightarrow \infty} - \frac{\log(x^{k-1} e^{-x/\theta})}{x}
 \end{aligned}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x} = 1/\theta.$$



$$\lim_{x \rightarrow \infty} \frac{\Gamma(s, x)}{x^{s-1} e^{-x}} = 1.$$







$$\begin{aligned}
&= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = - \lim_{x \rightarrow \infty} \frac{\log \operatorname{erfc} \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)}{x} \stackrel{(b)}{\geq} - \lim_{x \rightarrow \infty} \frac{\log \left(\sqrt{\frac{e}{2\pi}} e^{-2 \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)^2} \right)}{x}
\end{aligned}$$

$$\geq -\lim_{x \rightarrow \infty} \frac{\log \left(e^{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2} \right)}{x} = -\lim_{x \rightarrow \infty} \frac{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2}{x} = \frac{1}{\sigma^2} \lim_{x \rightarrow \infty} \frac{x^2 - 2x\mu + \mu^2}{x} = \infty,$$



$$\phi(x) \geq \sqrt{\frac{2e}{\pi}} \frac{\sqrt{\beta-1}}{\beta} e^{-\beta x^2},$$















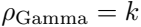










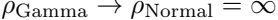














Q1 2024

2023年12月21日



$$\mathbb{E}[(S_1 + S_2)^2] = \int_0^1 (\cos(2\pi t) + \cos(2\pi t + \pi))^2 dt = \int_0^1 0 dt = 0,$$



1992 + 1992

$$= \int_0^1 \cos^2(2\pi t) dt + \int_0^1 \cos^2(2\pi t + \pi) dt$$

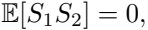
$$= \int_0^1 2 \cos^2(2\pi t) dt = \int_0^1 \cos(4\pi t) dt + 1 = 1.$$





$$E[S_1^2 + S_2^2] = E[S_1^2] + E[S_2^2] = E[S_1^2] + 2E[S_1 S_2] + E[S_2^2]$$

1991 + 1991 + 1991



$$\mathbb{E}[S_1 S_2] = \int_0^1 \cos(2\pi t) \cos(2\pi t + \pi) dt = \int_0^1 -\cos^2(2\pi t) dt = -\frac{1}{2} \int_0^1 \cos(4\pi t) dt - \frac{1}{2} = -\frac{1}{2},$$

01

,

02

E

1

0

,

2

1

1

2024 + 2025

2023-04-20

$$\mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1^2 + S_2^2]] = \mathbb{E} \phi_1 [\mathbb{E} [S_1^2]] + \mathbb{E} \phi_2 [\mathbb{E} [S_2^2]] + \mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1 S_2]]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2$$

$$+ \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} \int_0^1 \cos(2\pi t + \phi_1) \cos(2\pi t + \phi_2) dt d\phi_1 d\phi_2$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2 = 1/2 + 1/2 = 1.$$

1992



212122222





$${}_3F_2(1, 1, 1 + b; 2, 2; x) = \sum_{n=0}^{\infty} \frac{(1)_n (1)_n (1 + b)_n}{(2)_n (2)_n} \frac{x^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(1+b)_n}{(n+1)^2 n!} x^n = \frac{1}{b!} \sum_{n=0}^{\infty} \frac{(n+1)_b}{(n+1)^2} x^n$$

$$\underline{\underline{(a)}} \quad \frac{1}{b!} \sum_{n=0}^{\infty} \frac{\sum_{k=1}^b \begin{bmatrix} b \\ k \end{bmatrix} (n+1)^k}{(n+1)^2} x^n$$

$$\begin{aligned}
 &= \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \sum_{n=0}^{\infty} \frac{x^n}{(n+1)^{2-k}} \stackrel{(b)}{=} \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \frac{\operatorname{Li}_{2-k}(x)}{x},
 \end{aligned}$$

b

1

